Galois Cohomology

— *EYNTKA* Series —

Nathanael Chwojko-Srawley

September 19, 2025

Contents

Profinite Groups

In this section, a brief introduction to profinite groups is given. This is not an in-depth coverage, but rather it gives enough information to describe infinite galois theory.

Definition 0.0.1: Profinite Group

Take an inverse system of finite groups with the discrete topology. Then a *profinite group* is a topological group that is the limit of the system. Given any topological group G, we can take its *profinite completion* by taking:

$$\widehat{G} = \varprojlim_N G/N$$

where N ranges over all finite index open normal subgroups ordered by containment.

Any group G can be given a topology by given it the profinite topology given by taking the cosets of finite-index normal subgroups as a basis. The reader should check that if G is a topological group and $\widehat{G}$ is its profinite completion, then the continuous homomorphism $\varphi : G \to \widehat{G}$ can be shown that G is dense in $\widehat{G}$.

Example 0.0.2: Profinite Groups

1. Let G be finite. Then all normal subgroups have finite index and there are finitely many. It follows that $G \cong \widehat{G}$.

2. Let $G = \mathbb{Z}$. Then
$$\widehat{Z} = \varprojlim_n \mathbb{Z}/n\mathbb{Z} \cong \prod_p \mathbb{Z}_p$$
where $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$ is the p-adic completion of $\mathbb{Z}$.

3. Take $G = \mathbb{Q}$. Then $\mathbb{Q}$ has only one finite-index normal subgroup, namely itself. Hence $\widehat{\mathbb{Q}} = \{1\}$.

The above examples show that the map $\varphi : G \to \widehat{G}$ could be injective, surjective, both. The map will be injective if the intersections of all finite-index open normal subgroups of G is trivial (where we would then say G is residually finite). The completion comes equipped with a natural universal property where a continuous homomorphism $\varphi : G \to H$ with H a profinite group has a unique continuous homomorphism such that the following diagram commutes:

$$\begin{tikzcd} G \arrow[r, "\phi"] \arrow[dr, "\varphi"'] & \widehat{G} \arrow[d, "\exists!"] \\ & H \end{tikzcd}$$

A word on strongly complete? MIT notes p.275 Remark 26.15

Theorem 0.0.3: Characterizing Profinite Groups

A topological group is profintie if and only if it is a totally disconnected compact group

***Proof*:**
exercise: Compactness comes from Tychnoff's Theorem, while totally disconnected shall come from the discreteness in each finite quotient

Corollary 0.0.4: Distinguishing Profinite Completion

Let G be a profinite group. Then G is naturally isomorphic to its profinite completion, or even stronger:

$$G \cong \varprojlim G/U$$

where U ranges over all open normal subgroups ordered by contain need.
If G is isomorphic to its profinite completion *as a group* (i.e. is strongly complete) if and only if every finite index subgroup is open

***Proof*:**
MIT notes p.276

0.1 Group Cohomology

We start with no link to algebraic topology, and shall include the relation to singular cohomology before the examples.

Given a (not necessarily abelian) group G, what would it mean to extract cohomological information from it? A (co)chain would have to naturally be formed, and a functor must be found to derive. Groups themselves don't form an abelian category, but it was shown that representations, that is $k[G]$-modules, do form them. This is perhaps a first place in which we may think we can study groups, however this category is too nice (think of (co)homology with $\mathbb{Z}$ vs. $\mathbb{R}/\mathbb{C}$ coefficients). What is more useful is it's generalization:

Definition 0.1.1: G-Modules

Let G be an arbitrary group, and M an abelian group. Then M is a G-*module* if there exists a [left] group-action on M. Equivalently, if there exists a $\mathbb{Z}[G]$-module structure on M.

The reader is free to parse through the technicalities of the group action. All $k[G]$-modules presented in chapter **??** are $\mathbb{Z}[G]$-modules with the action restricted. An example of a G-module that is not a representation is the action of the group $\mathrm{Gal}_K(L)$ on the module L/K (a galois extension of L over K). All finite groups G have a non-trivial G-module given by the action of G on the group-ring $k[G]$ (note the addition is abelian). From a geometric perspective, modules represent sheaves (or in special cases vector bundles) over some schemes,

As we have seen many times at this point, the object acts on an abelian group can give lots of information on the object acting on it and the object being acted on. Given a G-module M, we can define the following:

Definition 0.1.2: G-Invariant subgroup

Let M be a G-module. Then the G-*invariant* set of M is the set:

$$M^G := \{m \in M \mid g \cdot m = m, \ \forall g \in G\}$$

which is the largest trivial G-submodule of M.

It is easy to see that $(-)^G$ is a functor. The following will help in computing cohomology

Corollary 0.1.3: Hom Representation of G-invariant

$$M^G \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, M^G) \cong \mathrm{Hom}_G(\mathbb{Z}, M)$$

***Proof*:**
exercise (what G-action should be put on $\mathbb{Z}$?)

Similarly, we can define:

Definition 0.1.4: G-Coinvariant subgroup

Let M be a G-module. Then the G-*coinvariant* of M is the set:

$$M_G := \frac{M}{\langle (gm - m) \mid g \in G, m \in M \rangle}$$

and is the maximal invariant quotient module with the trivial action.

Just like $(-)^G$, $(-)_G$ is also a functor. The reader should further verify that

Proposition 0.1.5: Tensor Representation of Coinvariant Module

$$M_G \cong_G \mathbb{Z} \otimes_{\mathbb{Z}[G]} M$$

Proposition 0.1.6: Invariants Is Left-exact

The functor $(-)^G$ is left-exact, the functor $(-)_G$ is right exact.

***Proof*:**
First, prove that $(-)^G$ is right-adjoint to the trivial-action functor.

$$\mathrm{Hom}(T(M), N) \cong \mathrm{Hom}(M, N^G)$$

And similarly for $(-)_G$; it is left-adjoint to the trivial-action functor. As $T(-)$ is an exact functor, $(-)^G$ and $(-)_G$ are left-exact and right-exact respectively, completing the proof.

Both of these functors are not exact:

Example 0.1.7: Lack Of Exactness's
Let $G = C_2$ be the cyclic group of order 2 and:

$$0 \to I_G \to \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where $\epsilon(1 \cdot a + \sigma \cdot b) = a + b$ and I_G is the kernel. This sequence is exact. Take $(-)^G$:

$$0 \to (I_G)^G \to (\mathbb{Z}[G])^G \xrightarrow{\epsilon^G} (\mathbb{Z})^G \to 0$$

i.e.:

$$0 \to 0 \to \Delta \xrightarrow{\epsilon^G} \mathbb{Z} \to 0$$

where $\Delta = \{(a, a) \mid a \in \mathbb{Z}\}$. Then ϵ^G is no longer surjective as $\epsilon^G(a, a) = 2a$. The above example can also be used to show $(-)_G$ is not left-exact. More generally, if $0 \to M_2^G \to M_2 \to M_3 \to 0$ is exact, taking $(-)^G$ doesn't change $M - 2^G$, but now we get an isomorphism between the 1st and second terms, so it cannot be surjective.

More generally, let G act trivially on a coset $[m]$ of a quotient M/L. Can this action always be lifted so that G acts trivially on a representative m?

Thus, it is fruitful to take the ith right-derived functor of $(-)^G$ (similarly the left-derived functors of $(-)_G$):

$$\mathbf{R}^i(-)^G := H^i(G, -)$$

which would give us the long exact sequence of abelian groups:

$$0 \to L^G \to M^G \to N^G \to H^1(G, L) \to H^1(G, M) \to \cdots$$

Since $M^G = \mathrm{Hom}_G(\mathbb{Z}, M)$ by taking an injective resolution of M, or a projective resolution of $\mathbb{Z}$, we can write:

$$H^i(G, M) = \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, M)$$

Note that $H^0(G, M) = M^G$, hence the higher order cohomology groups represent some failure for points to be fixed. Let us compute some examples to show the power of the machinery developed in chapter **??** before concretely demonstrating how the cohomology groups represent the obstruction of points being fixed.

Example 0.1.8: Group Cohomologies – Homological Algebra Computations

1. The trivial group has trivial cohomology
2. (Integers) Let $G = \mathbb{Z}$. Then $\mathbb{Z}[G] \cong \mathbb{Z}[x, x^{-1}]$. Then since $\mathbb{Z}[x, x^{-1}]/(1-x) \cong_G \mathbb{Z}$ with the trivial action, define the following free (hence projective) resolution:

$$\cdots \to 0 \to \mathbb{Z}[x, x^{-1}] \xrightarrow{\cdot(1-x)} \mathbb{Z}[x, x^{-1}] \to 0 \to \cdots$$

   Applying the (contravariant) $\mathrm{Hom}_{\mathbb{Z}[x,x^{-1}]}(-, M)$, we can compute the cohomology of

$$\cdots \to 0 \to M \xrightarrow{\cdot(1-x)} M \to 0 \to \cdots$$

   Then we can conclude by direct computation:

$$H^0(\mathbb{Z}, M) \cong \ker(M \xrightarrow{(\cdot(1-x))} M) = M^{\mathbb{Z}}$$

   and:

$$H^1(\mathbb{Z}, M) = \frac{M}{(1-x)M}$$

   and furthermore $H^i(\mathbb{Z}, M) = 0$ for $i \notin \{0, 1\}$
3. Let $G = C_n$ be a finite cyclic group of over n with generator σ. Let us form a free resolution of $\mathbb{Z}[G]$ by taking advantage of the fact that $\sigma - 1$ and $N = \sum_{i=0}^{n-1} \sigma^i$ are zero divisors:

$$N \cdot (\sigma - 1) = (1 + \sigma + \cdots + \sigma^{n-1})(\sigma - 1) = \sigma^n - 1 = 1 - 1 = 0$$

   This gives a resolution:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon}\to \mathbb{Z} \to 0$$

   where

$$\epsilon\left(\sum_g a_g g\right) = \sum_g a_g$$

   is the usual augmentation map. Then:

$$H^k(C_n, M) = \begin{cases} \frac{M^G}{NM} & k \text{ even, } k \geq 2 \\ \frac{{}_N M}{(\sigma-1)M} & k \text{ odd, } k \geq 1 \end{cases} \qquad \text{where} \qquad {}_N M = \{m \in M \mid N \cdot m = 0\}$$

   For example, if $M = \mathbb{Z}$, then:

$$H^k(C_n, \mathbb{Z}) = \begin{cases} \mathbb{Z}/m\mathbb{Z} & k \text{ even, } k \geq 2 \\ 0 & \text{otherwise} \end{cases}$$

4. Let $G = *_{s\in S}\mathbb{Z}$ be the free product of S copies of $\mathbb{Z}$. This has a simple free resolution, namely take:
$$\epsilon : \mathbb{Z}[G] \to \mathbb{Z}$$
where the kernel is:
$$I_S = \sum_{s\in S} \mathbb{Z}[G] \cdot (s-1)$$
It should be verified that I_S is a free G-module, hence we have a free-resoluion:
$$0 \to I_S \to \mathbb{Z}[G] \xrightarrow{\epsilon}\!\!\to \mathbb{Z} \to 0$$
and cohomology:
$$H^k(*_{s\in S}\mathbb{Z}, \mathbb{Z}^G) = \begin{cases} \mathbb{Z} & k=0 \\ \bigoplus_{s\in S}\mathbb{Z} & k=1 \\ 0 & k\geq 2 \end{cases}$$

5. (If you know some algebraic geometry) *Galois Cohomology*: Take the category of varieties W over k such that when extending scalars to $\overline{k}$ we have $V/\overline{k} \cong W/\overline{k}$. Let's label it $\mathbf{Var}_{k\to\overline{k}}$, and say that the objects are:
$$\mathrm{Obj}(\mathbf{Var}_{k\to\overline{k}}) = \left\{W/k \;\middle|\; V/\overline{k} \cong W/\overline{k}\right\}$$
and the morphism are regular morphisms. Then I claim that:
$$\mathbf{Var}_{k\to\overline{k}} \cong H^1(\mathrm{Gal}_k(\overline{k}), \mathrm{Aut}(V/\overline{k}))$$
To see this, first note that an element $\sigma \in \mathrm{Gal}_k(\overline{k})$ acts on a $\overline{k}$-variety $W/\overline{k} = W_k \times_{\mathrm{Spec}\,(k)} \mathrm{Spec}\,(\overline{k})$ via:
$$\mathrm{id}_{W_k} \times \mathrm{Spec}\,(\sigma^{-1}) : W/\overline{k} \to W/\overline{k}$$
Or, for those less comfortable with fibered produced from algebraic geometry, they can think of this map as the "equivalent":
$$\mathrm{id}_{W_k} \otimes_k \sigma^{-1} : k[W] \otimes_k \overline{k}$$
With the action defined, let us look at the correspondence. Take $W_k \in \mathbf{Var}_{k\to\overline{k}}$ so that $\varphi : W_{\overline{k}} \xrightarrow{\sim} V_{\overline{k}}$ is an isomorphism. Then notice that this isomorphism induces a map $\varphi \circ \sigma^{-1} \circ \varphi^{-1}$, which is an automorphism on $W_{\overline{k}}$. The collection of all these maps as σ varies can uniquely define W_k (k is the fixed subfield over all σ). This is true up to a choice of $\mathrm{Aut}_k(V)$, and so we quotient out by it. Then we can see this creates the correspondence between these two categories, and shows how far away the category $\mathbf{Var}_{k\to\overline{k}}$ is from being invariant to the group-action of the galois group.

6. (if you know elliptic curves) An excellent example of where galois cohomology is immediately useful is in the theory of elliptic curves. On a high level, it is relatively easy to find the solution of an elliptic curve over $\mathbb{C}$, but it becomes much more difficult to solve it over finite extensions of $\mathbb{Q}$, or $\mathbb{Q}$ itself. If the reader has read (or is reading) [1], then the have encountered the short exact sequence of $\mathrm{Gal}_K(\overline{K})$-modules:
$$0 \to E[m] \to E(\overline{K}) \xrightarrow{[m]} E(\overline{K}) \to 0$$

where $E[m]$ are the m-torsion points of an elliptic curve ($m \geq 2$), K is a number field, $E(\overline{K})$ are the points of the elliptic curve over $\overline{K}$. Let $G = \mathrm{Gal}_K(\overline{K})$. Then when considering solutions over K, the short exact sequence sequence becomes a long exact sequences:

$$0 \to E(K)[m] \to E(K) \xrightarrow{[m]} E(K) \xrightarrow{\delta} H^1(G, E[m]) \to H^1(G, E(\overline{K})) \xrightarrow{[m]} H^1(G, E(\overline{K}) \to \cdots$$

This is the foundation on which the torsion-free points of E/K will be computed!

7. If the reader knows about the classifying space BG, then:

$$H^n(BG, \mathbb{Z}) \cong H^n(G, \mathbb{Z})$$

where he left hand side is singular cohomology and the right hand side is group cohomology.

What obstruction information does this capture? Since there is a failure in surjectivity $\varphi : M \twoheadrightarrow N$ when applying $(-)^G$, this means that a fixed element $n \in N^G$ does *not* necessarily come form a fixed element $m \in M^G$. Thus, even though $\varphi(m) = n$ exists, it need not be the case that there exists an $m' \in M^G$ such that $\varphi^G(m') = n$. To measure this failure, take the short exact sequence

$$0 \to L \xrightarrow{\iota} M \xrightarrow{\varphi} N \to 0$$

and let $n \in N^G$. As $N^G \subseteq N$, there exists an $m \in M$ such that $\varphi(m) = n$. Consider any $g \in G$ and take $g \cdot m - m$. Then applying the G-homomorphism:

$$\varphi(g \cdot m - m) = g \cdot \varphi(m) - \varphi(m) = g \cdot n - n \overset{!}{=} n - n = 0$$

where $\overset{!}{=}$ comes from $n \in N^G$. By exactness $\ker \varphi = \mathrm{im} \iota$, so there must exist an element $\lambda_m(g) \in L$ such that

$$\iota(\lambda_m(g)) = g \cdot m - m$$

Now, $n \in N^G$ can be lifted if and only if $g \cdot m - m = 0$ for all $g \in G$, since then $m \in M^G$. In terms of elements of L:

$$\forall g \in G, \ \iota(\lambda_m(g)) = 0$$

by injectivity, it must be that $\lambda_m(g) = 0$ for all $g \in G$. Thus, the function $\lambda_m : G \to L$ is non-trivial if the lift does not exist. Let us explore the properties of λ_m. First, if $g_1, g_2 \in G$ then:

$$\begin{aligned}
\iota(\lambda_m(g_1 g_2)) &= g_1 g_2 \cdot m - m \\
&= g_1 g_2 \cdot m - g_1 \cdot m + g_1 \cdot m - m \\
&= g_1 \cdot (g_2 \cdot m - m) + (g_1 \cdot m - m) \\
&= g_1 \iota(\lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1 \lambda_m(g_2)) + \iota(\lambda_m(g_1)) \\
&= \iota(g_1 \lambda_m(g_2) + \lambda_m(g_1))
\end{aligned}$$

Thus, as f is injective (more specifically a monomorphism) we get:

$$\lambda_m(g_1 g_2) = \lambda_m(g_1) + g_1 \lambda_m(g_2)$$

A function with this property is given a name:

Definition 0.1.9: 1-Cocycle

A function $f : G \to M$ is a called a *1-cocycle*, *1-derivation*, or *crossed homomorphism* if:

$$f(gh) = f(g) + gf(h)$$

The set of all derivations is denoted $Z^1(G, M)$. If it is clear from context, the "1-" is dropped.

Now, the above λ_m was dependent on a choice of m. But $M \hookrightarrow N$ is surjective, and hence there may be another m' such that $\varphi(m') = n$. As φ is a G-homomorphism

$$\varphi(m - m') = c - c = 0$$

and so $m - m' \in \ker \varphi = \mathrm{im} \iota$. Thus, there is an $\ell \in L$ such that

$$m - m' = \iota(\ell) \qquad \text{or} \qquad m' = m + \iota(\ell)$$

Thus, let us relate $\lambda_{m'}$ to λ_m:

$$\begin{aligned}
\iota(\lambda_{m'}(g)) &= g \cdot m' - m' \\
&= g \cdot (m + \iota(\ell)) - m - \iota(\ell) \\
&= g \cdot m + g \cdot \iota(\ell)) - m - \iota(\ell) \\
&= (g \cdot m - m) + g \cdot \iota(\ell) - \iota(\ell) \\
&= \iota(\lambda_m(g)) + \iota(g \cdot \ell) - \iota(\ell) \\
&= \iota(\lambda_m(g) + g \cdot \ell - \iota(\ell))
\end{aligned}$$

Thus, as ι is a monomorphism:

$$\lambda_{m'}(g) = \lambda_m(g) + g \cdot \ell - \ell$$

Thus, the term $g \cdot \ell - \ell$ can be seen as the "trivial" part of the obstruction, as it does not change the difficulty in finding an obstruction. This is given a name:

Definition 0.1.10: 1-Coboundary

A function $f_m : G \to M$ is a called a *1-coboundary* or *principal derivation* if:

$$f_m(g) = g \cdot m - m$$

The set of all boundary maps is denoted $B^1(G, M)$.If it is clear from context, the "1-" is dropped.

Note that a coboundary is a cocycle:

$$f_m(gh) = gh \cdot m - m = gh \cdot m - h \cdot m + h \cdot m - m = f_m(g) + gf_m(h)$$

Hence $B^1(G, M) \subseteq Z^1(G, M)$. It shall be shown in a moment that:

$$H^1(G, M) \cong \frac{Z^1(G, M)}{B^1(G, M)}$$

First, let us show there is a map $L^G \to H^1(G, M)$. Indeed there is a well-defined G-module homomorphism:

$$\ell \mapsto [g \mapsto g \cdot m - m]$$

where m is in the fiber of ℓ. Let us next motivate the 2nd order cohomology group; let us assume the cocycle-representation and the derived functor representation are isomorphic for the next part. Let us start with the long exact sequence we know is given:

$$0 \to L^G \xrightarrow{\alpha} M^G \xrightarrow{\beta} N^G \xrightarrow{\delta^0} H^1(G, L) \xrightarrow{\alpha_*} H^1(G, M) \xrightarrow{\beta_*} H^1(G, N) \xrightarrow{\delta^1} H^2(G, L) \cdots$$

Consider $[\nu] \in H^1(G, N)$; let us call it a 1-obstruction. Then can it be lifted some $[\mu] \in H^1(G, M)$, namely does there exist $[\mu]$ such that $\beta_*([\mu]) = [\nu]$. To find out, first take some representative cocycle $\nu : G \to N$ (notice we are now working with a map, not an element). As $M \hookrightarrow N$ is surjective, we have a diagram:

$$\begin{tikzcd} & M \arrow[d, "\beta"] \\ G \arrow[ur, dashed, "\mu?"] \arrow[r, "\nu"'] & N \end{tikzcd}$$

We can certainly lift each point *locally*, namely given a point $\nu(g) = n \in N$ there is a point (or points) m such that $\beta(m) = n$[1]. But this lifting can fail *globally*, namely such a lift is not guaranteed to be a 1-cocycle. This mirrors the general pattern in cohomology where having local data of a function does not imply the existence of a global function that restricts to all the local data. Let us measure this failure of lifting. If the lifted μ is a cocycle, it would satisfy:

$$\mu(g_1 g_2) - \mu(g_1) - g_1\mu(g_2) = 0$$

Thus, define the function:

$$\psi : G \times G \to M, \qquad \psi(g_1, g_2) = \mu(g_1 g_2) - \mu(g_1) - g_1\mu(g_2)$$

Let us show that for any $(g_1, g_2) \in G \times G$, $\psi(g_1, g_2) \in \ker \beta$. Indeed:

$$\begin{aligned} \beta(\psi(g_1, g_2)) &= \beta(\mu(g_1 g_2) - \mu(g_1) - g_1\mu(g_2) \\ &= \beta(\mu(g_1 g_2)) - \beta(\mu(g_1)) - \beta(g_1\mu(g_2)) \\ &\overset{!}{=} \nu(g_1 g_2)) - \nu(g_1) - g_1\nu(g_2)) \\ &= 0 && \nu \text{ is a cocycle} \end{aligned}$$

where $\overset{!}{=}$ comes from μ being defined as a lift of ν and hence $\beta \circ \mu = \nu$. Thus, by exactness $\ker \beta = \mathrm{im}\alpha$ and so for each pair $(g_1, g_2) \in G \times G$, there exists a unique element in L that maps to $\psi(g_1, g_2)$, namely there exists a function $\Omega : G \times G \to L$ such that

$$\alpha(\Omega(g_1, g_2)) = \psi(g_1, g_2)$$

This Ω would now be our new 2-cocycle and are denoted $Z^2(G, M)$. Doing a similar but longer computations such as the one for the 1-cycle, we can find that:

$$g_1\Omega(g_2, g_3) - \Omega(g_1 g_2, g_3) + \Omega(g_1, g_2 g_3) - \Omega(g_1, g_2) = 0$$

[1]Technically, the axiom of choice is invoked here for making a potentially uncountable number of choices

Similarly, an explicit computation would fine that 2-boundaries (given a 1-chain ψ) are of the form

$$\Omega(g_1, g_2) = g_1 \cdot \psi(g_2) - \psi(g_1 g_2) + \psi(g_1)$$

These are label $B^2(G, M)$ and:

$$H^2(G, M) \cong \frac{Z^2(G, M)}{B^2(G, M)}$$

The same process continues for higher order cohomology. For the reader who knows some algebraic topology, the following informal discussion grounds the higher order group-cohomology as the singular (or simplicial) cohomology of BG, the classifying space of G with the property that $\pi_1(BG) = G$ and $\pi_n(BG) = \{0\}$ for all $n > 1$ (see [2]). In particular:

$$H^n_{\text{Sing}}(BG, M) \cong H^n(G, M)$$

Note that if M has a non-trivial G-module action, it corresponds to cohomology with local coefficients, but the geometric picture remains the same. Using the nerve or bar construction on BG and considering the usual covering $EG \to BG$ we get the following correspondence:

Geometric Object in EG	Correspondence in BG	Algebraic equivalent
0-Simplices (Vertices)	A single 0-simplex, base point $*$.	elements of G
1-Simplices (Edges)	Directed edge for each element $g \in G$. An arrow from $*$ to $*$ labeled by g.	Functions on G
2-Simplices (Triangles)	Oriented triangle for each pair $(g_1, g_2) \in G \times G$. Edges are the 1-simplices for g_1, g_2, and the product $g_1 g_2$.	Functions on $G \times G$
3-Simplices (Tetrahedral)	Oriented tetrahedron for each triple $(g_1, g_2, g_3) \in G \times G \times G$. Faces are the triangles for (g_2, g_3), $(g_1, g_2 g_3)$, etc.	Functions on $G \times G \times G$
n-Simplices	n-tuple $(g_1, \ldots, g_n) \in G^n$.	Functions on G^n

Where "functions" here means set-function (to take into account the "failed" lifting), and the transition maps d_i captures the G-module homomorphism information. Geometrically, the G-module homomorphisms correspond to G-equivariant maps $f : G^n \to M$.

Using this intuition, we can construct a standard way of defining a projective resolution of $\mathbb{Z}$ by G-modules. First, let $\mathbb{Z}[G^n]$ be a G-module with the diagonal action given by multiplying on the left, or more specifically:

$$g \cdot (g_1, ..., g_n) = (gg_1, ..., gg_n)$$

Note that this diagonal map corresponds to the natural action of left multiplication by g on EG. These modules are free G-modules; clearly $\mathbb{Z}[G]$ is a $\mathbb{Z}$-module with basis given by the elements G. Then for $n \geq 0$:

$$\mathbb{Z}[G^{n+1}] \cong \bigoplus_{(g_1, ..., g_n) \in G^n} \mathbb{Z}[G] \cdot (1, g_1, ..., g_n)$$

which shows that these are free. From this, we can consider the following projective (even free) resolution:

Definition 0.1.11: Standard Resolution

Let G be a group. Then the *standard resolution* of $\mathbb{Z}$ by G-modules is the exact sequence of G-module homomorphism:

$$\cdots \to \mathbb{Z}[G^{n+1}] \xrightarrow{d_{n+1}} \mathbb{Z}[G^n] \xrightarrow{d_n} \cdots \xrightarrow{d_2} Z[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where on the group G we define:

$$d_n(g_0, ..., g_n) = \sum_{i=1}^{n} (-1)^i (g_0, ..., \hat{g}_i, ..., g_n)$$

where the term $\hat{g}_i$ is omitted from the tuple. The map ϵ is also called the *augmentation map*.

Note that each $\mathbb{Z}[G^{n+1}]$ is a We claimed this is an exact sequence, however this ought to be verified.

Lemma 0.1.12: Standard Resolution is Exact

The standard resolution of $\mathbb{Z}$ by G-modules is exact

This shows that the nomenclature of cocycles and coboundaries above is used correctly.

> ***Proof*:**
> This a proof that is about keeping track a lot of technical details, but it is all eminently doable, and so left to the willing reader.

Hence, after taking the contravariant functor $\mathrm{Hom}_{\mathbb{Z}[G]}(-, M)$, we get the cochain complex:

$$0 \to \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G], M) \to \cdots \to \mathrm{Hom}_{\mathbb{Z}[G^n]}(-, M) \xrightarrow{d_n^*} \mathrm{Hom}_{\mathbb{Z}[G^{n+1}]}(-, M) \to \cdots$$

where d_n^* maps $\varphi \mapsto \varphi \circ d_n$. Similarly, we can take the covariant functor

$$\cdots \to \mathbb{Z}[G^3] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{2*}} \mathbb{Z}[G^2] \otimes_{\mathbb{Z}[G]} A \xrightarrow{d_{1*}} \mathbb{Z}[G] \otimes_{\mathbb{Z}[G]} A \xrightarrow{\epsilon} 0$$

where d_{n*} is given by $(g_0, ..., g_n) \otimes a \mapsto d_n(g_0, ..., g_n) \otimes a$, the map ϵ is the augmentation map given by $\sum a_g g \mapsto \sum a_g$, and we view the modules as *right* $\mathbb{Z}[G]$-modules wit the action

$$(g_1, ..., g_n) \cdot g = (g_1 g, ..., g_n g)$$

Let us now better understand these hom-sets and relate to the above discussion. Let

Definition 0.1.13: coChain

Let
$$C^n(G, M) = (\mathrm{Hom}_{\mathbf{Set}}(G^n, M), +)$$
with group operation defined on the maps given by point-wise addition. This group is called the *n-cochain group*.

From this, we can define a boundary map:

$$d^n : C^n(G, M) \to C^{n+1}(G, M)$$
$$d^n(f)(g_0, ..., g_n) = g_0 f(g_1, ..., g_n) - f(g_0 g_1, g_2, ..., g_n)$$
$$+ f(g_0, g_1 g_2, ..., g_n) - \cdots + (-1)^n f(g_0, ..., g_{n-2}, g_{n-1} g_n) + (-1)^{n+1} f(g_0, ..., g_{n-1})$$

Then $d^{n+1} \circ d^n = 0$, giving us a cochain complex. We can then have:

Definition 0.1.14: Cohomology Group – using Cocycles and Coboundaries

$$H^n(G, M) = \frac{\ker d^n}{\mathrm{im} d^{n-1}} = \frac{Z^n(G, M)}{B^n(G, M)}$$

We sum up many of the results that we've shown before using the new notation:

1. $C^0(G, M) = M$, and $H^0(G, M) \cong M^G$
2. $d^0 : C^0(G, M) \to C^1(G, M)$ is given by $d^0(a)(g) = ga - a$
3. $H^0(G, M) \cong \ker d^0 = M^G$
4. $\mathrm{im} d^0 = \{f : G \to M \mid \exists a \in M$ such that $f(g) = ga - a, \ \forall g \in G\}$ (principal crossed homomorphism)
5. $d^1 : C^1(G, M) \to C^2(G, M)$ given by $d^1(f)(g, h) = gf(h) - f(gh) - f(g)$
6. $\ker d^1 = \{f : G \to M \mid f(gh) = f(g) + gf(h)\}$ (crossed homomorphisms)
7. If $0 \to A \to B \to C \to 0$ is exact, we get an exact sequence of cochains for every n:
$$0 \to C^n(G, A) \to C^n(G, B) \to C^n(G, C) \to 0$$

We shall show that the collection of complexes given by the above is isomorphic to those given by the standard resolution

Proposition 0.1.15: Characterizing Hom of Resolution

Let M be a G-module. Then for every $n \geq 0$, we there is an isomorphism:

$$\Phi^n : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^{n+1}], M) \xrightarrow{\cong} C^n(G, M)$$

where the map $\varphi : \mathbb{Z}[G^{n+1}] \to M$ goes to $f : G^n \to M$ given by:

$$f(g_1, ..., g_n) = \varphi(1, g_1, g_1 g_2, g_1 g_2 g_3, ..., g_1 g_2 g_3 \cdots g_n)$$

The isomorphism Φ^n commutes with boundary maps, that is

$$\Phi^{n+1} \circ d^*_{n+1} = d^n \circ \Phi^n$$

Hence we have an isomorphism from the category of cochain complexes with the standard resolution to the one with boundary-cycle maps.

***Proof*:**
very doable, and worth doing at least once.

Due to the relevance of the standard representation, we have the following definition:

Definition 0.1.16: Induced and Coinduced G-module

Let G be a group and A an abelian group. Then:

$$\mathrm{CoInd}^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$$

with the G-action defined as $(g\varphi)(z) = \varphi(zg)$ is called the *coinduced G-module* associated to A. Similarly

$$\mathrm{Ind}^G(A) = \mathbb{Z}[G] \otimes_{\mathbb{Z}} A$$

with G-action defined by $g(z \otimes a) = (gz) \otimes a$ is the is the induced G-module associated to A.

A result that is common in many Cohomologies theories is some version of the following:

Lemma 0.1.17: Induced and Coinduced Equivalence

Let G be a finite group and A an abelian group. Then

$$\mathrm{Ind}^G(A) \cong \mathrm{CoInd}^G(A)$$

As a consequence, if $B = \mathrm{Ind}^G(A)$ or $B = \mathrm{CoInd}^G(A)$, then

$$H_0(G, B) \cong H^0(G, B) \cong A$$

and they are both zero for $n \geq 1$.

***Proof*:**
Define the map: $\alpha : \mathrm{Coind}^G(A) \to \mathrm{Ind}(G)(A)$ given by:

$$\varphi \mapsto \sum_{g \in G} g^{-1} \otimes \varphi(g)$$
$$(g^{-1} \mapsto a) \leftarrow\!\shortmid g \otimes a$$

with $(g^{-1} \mapsto a)(h) = 0$ for $h \in G \setminus \{g^{-1}\}$. These are certainly inverses, and easily verifiable to be G-module homomorphisms (this is a similar technique used in Maschkle's Theorem (see **??**).

The last part is left for exercise ref:HERE.

Proposition 0.1.18: Torsion of H^1

if G has order n and is finite, then every element of $H^1(G, M)$ has order divisible by n.

***Proof*:**
Let $\delta \in Z^1$ so that $\delta(gh) = \delta(g) + g\delta(h)$ for all $g \in G$. Sum both sides over $h \in G$ keeping g fixed. Then as multiplying by g is translation:

$$\sum_{h \in G} \delta(gh) = \sum_{h \in G} \delta(h) = S$$

As $\delta(g)$ is a constant, we can re-arrange and get:

$$(1 - g)S = |G|\delta(g)$$

Now, re-write it suggestively as:

$$g(-S) - (-S) = |G|\delta(g)$$

But now the left hand side is a coboundary, and hence the equation is 0 in $H^1(G, M)$ showing that the order of the group zero's any cocycle, completing the proof.

Proposition 0.1.19: M finitely generated, then H1 finite

Let M be a finitely generated G-module and G finite. Then $H^1(G, M)$ is finite.

***Proof*:**
Let $\langle m_1, ..., m_r \rangle = M$. Then C^1 is finitely generated as a $\mathbb{Z}$-module (note that G had to be finite). Then $Z^1(G, M)$ (as $\mathbb{Z}$-submodule) is finitely generated. But now by proposition 0.1.19, every element has order divisible by $n = |G|$. Thus, take the lcm of the order of the generators, giving a bound of the order of H^1, completing the proof.

Let us give some examples within galois cohomology. Note that taking G-invariance of a short exact sequence of G-modules defined using some field L/K will reduce to the field L to the field K, as shall be evident soon. Let us start with the infamous Hilbert's 90th theorem. Let L/K be a cyclic extension so that $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$. The G acts on $L^\times$ in the natural way, and hence we find it's group cohomology. First:

Lemma 0.1.20: Cocycles using Norm and Trace

Let L/K be a cyclic extension $G = \mathrm{Gal}_K(L) \cong \mathbb{Z}/m\mathbb{Z}$ with generator σ, and $Z^1(G, L^\times)$ the 1-cocycles. Let $U_{L/K} = \left\{ l \in L \;\middle|\; \mathrm{Nm}_{L/K}(l) = 1 \right\}$ (sometimes this is denoted L^1). Then as groups:

$$\begin{aligned} Z^1(G, L^\times) &\cong U_{L/K} \\ c &\mapsto c(\sigma) \end{aligned}$$

Similarly:

$$Z^1(G, L) \cong T_{L/K}$$

where $T_{L/K}$ is the traceless elements[a]

[a] There is less common notation, sometimes it is denoted L^0

***Proof*:**
Let us first show the map is well-defined. Note first that a cocycle is determined by where it maps σ, namely for an arbitrary $c \in Z^1(G, L^\times)$:

$$\begin{aligned} c(\sigma^2) &= c(\sigma) \cdot \sigma(c(\sigma)) \\ c(\sigma^3) &= c(\sigma) \cdot \sigma(c(\sigma^2)) \\ &\vdots \end{aligned}$$

Next, note that $c(\sigma^m) = c(e) = 1$ since:

$$c(e) = c(e \cdot e) = c(e) \cdot e(\sigma(e)) = c(e) \cdot c(e)$$

as these map to $L^\times$, all elements are invertible, hence:

$$c(e) = 1 \in L^\times$$

Then:

$$1 = c(\sigma^n) = c(\sigma) \cdot \sigma(c(\sigma)) \cdot \sigma^2(c(\sigma)) \cdots \sigma^{n-1}(c(\sigma))$$

Letting $\beta = c(\sigma)$:

$$1 = c(\sigma^n) = \beta \cdot \sigma(\beta) \cdot \sigma^2(\beta) \cdots \sigma^{n-1}(\beta) = \prod_{g \in G} g(\beta) = \mathrm{Nm}_{L/K}(\beta)$$

showing the map is well-defined. As it is determined by where σ maps to, if $c_1, c_2 \in Z^1(G, L^\times)$ and $c_1(\sigma) = c_2(\sigma)$ then they agree on each σ^k and hence $c_1 = c_2$, showing injectivity. Surjectivity can be shown by defining $c(\sigma) = \beta$ for $\beta \in U_{L/K}$ and showing it satisfies the cocycle condition, which will be left as an exercise.

A similar argument can be done using the trace.

Theorem 0.1.21: Hilbert's 90th Theorem – Cohomology Upgrade

Using the above lemma's notation:

$$H^1(G, L^\times) \cong \{1\}$$

***Proof*:**
By Hilbert's 90th theorem (**??**), every norm 1 element is of the form:

$$\frac{\alpha}{\sigma^k(\alpha)}$$

But this is a coboundary element, hence all cocycles are coboundaries, completing the proof.

Note this theorem generalizes to finite galois extensions in general, see ref:HERE.

Let us use this to find some interesting results. Consider the short exact sequence:

$$1 \to \mu_n(L) \to L^\times \to (L^\times)^n \to 1$$

Taking G-invariance, we get:

$$1 \to \mu_n(K) \to K^\times \to (L^\times)^n \cap K$$

where we can't just put $(K^\times)^n$ (for example $-1 \in (\mathbb{Q}(i)^\times)^2 \cap \mathbb{Q}$ but $-1 \notin (\mathbb{Q}^\times)^2$). By cohomology we get a long exact sequence:

$$1 \to \mu_n(K) \to K^\times \to (L^\times)^n \cap K \to H^1(G, \mu_n(L)) \to H^1(G, L^\times) \to \cdots$$

By Hilbert's 90TH, $H^1(G, L^\times) = 1$ so

$$1 \to \mu_n(K) \to K^\times \to (L^\times)^n \cap K \to H^1(G, \mu_n(L)) \to 0$$

Thus, $(L^\times)^n \cap K \twoheadrightarrow H^1(G, \mu_n(L))$ with kernel $(K^\times)^n$, namely:

$$\frac{(L^\times)^n \cap K}{(K^\times)^n} \cong H^1(G, \mu_n(L))$$

If $\mu_n \subseteq K$ then this group-action is trivial on $\mu_n(L)$, and so:

$$H^1(G, \mu_n(L)) = \operatorname{Hom}(G, \mu_n(K))$$

where we are assuming $\mu_n(K) = \mu_n(L)$ (we don't want the extension to introduce new roots of unity). Now, as $\operatorname{Hom}(G, \mu_n(K))$ is abelian (as $\mu_n(K)$ is abelian), we get:

$$\operatorname{Hom}(G, \mu_n(K)) \cong \operatorname{Hom}\left(\frac{G}{[G,G]}\mu_n(K))\right)$$

This gives us a very easy condition where this group is trivial, namely if $|G/[G,G]|$ is coprime to n (remember only finite galois extensions are being considered for now), then

$$\operatorname{Hom}\left(\frac{G}{[G,G]}\mu_n(K))\right) = \{0\}$$

In which case:

$$(L^\times)^n \cap K \cong (K^\times)^n$$

For a simple example, take $n = 2$ (for ± 1 will always be in a field). Taking an extension of odd degree, then a square of an element in the above field L will be the square of an element in the bellow field K.

The next example comes from principal (fractional) ideals P_L. Consider:

$$1 \to \mathcal{O}_L^\times \to L^\times \to \mathcal{P}_L \to 1$$

Taking the $G = \mathrm{Gal}_K(L)$-invariance and extending to a long exact sequence while using Hilbert's 90th:

$$1 \to \mathcal{O}_K^\times \to K^\times \to \mathcal{P}_L^G \to H^1(G, \mathcal{O}_L^\times) \to 0$$

Thus using exactness and that the kernel of the map is $\mathcal{P}_K$ we get:

$$\frac{\mathcal{P}_L^G}{\mathcal{P}_K} \cong H^1(G, \mathcal{O}_L^\times)$$

This H^1 is often non-trivial. Consider for example $Q(\sqrt{-d})/\mathbb{Q}$ so that $G = \mathbb{Z}/2\mathbb{Z}$. Then $\mathcal{O}_L^\times$ is finite and:

$$|\mathcal{O}_L^\times| = \begin{cases} 4 & L = Q(i) \\ 3 & L = Q(\sqrt{-3}) \\ 2 & \text{otherwise} \end{cases}$$

Taking the generic case of there being two elements, G acts trivially and so:

$$H^1(G, \{\pm 1\}) \cong \mathrm{Hom}(\mathbb{Z}/2\mathbb{Z}, \mathbb{Z}/2\mathbb{Z}) \cong \mathbb{Z}/2\mathbb{Z}$$

showing that generically there is a non-trivial obstruction to the generators of the principal ideal P_L being a generator of an ideal from P_K.

Let us now address some functorial properties. If $f : M_1 \to M_2$ is a G-module homomorphism, then f can pushforward H^1, namely:

$$f_* : H^1(G, M_1) \to H^1(G, M_2) \qquad f_*([c]) = [f \circ c]$$

It is left as (simple) exercise to show it satisfies the co-cycle relation and is independent of the coboundary relation.

Next, consider $H \trianglelefteq G$. This defines two natural cohomologies:

Definition 0.1.22: Restriction and Inflation

Let M be a G-module and $H \trianglelefteq G$.

1. Restriction: $H^1(G, M) \to H^1(H, M)$ Given $G \to M$ we get $H \to G \to M$ and get cocycles from there.

2. Inflation: $H^1(G/H, M^H) \to H^1(G, M)$: given $G/H \to M^H$ we get

$$G \to G/H \to M^H \to M$$

   and we get cocycles from there.

Proposition 0.1.23: Hochschild Exact Sequence

There is an exact sequence:

$$1 \to H^1(G/H, M^H) \to H^1(G, M) \to H^1(H, M)$$

***Proof*:**
Let $\delta \in H^1(G/H, M^H)$, and $\overline{\delta} : G \to M$ be the image under inflation. Let's show the inflation map is an injection. If $\overline{\delta}$ is a 1-coboundary, so there exits $m \in m$ so that $\overline{\delta} = gm - m$. On the other hand, $\overline{\delta}$ arose from the composition in the definition:

$$G \to G/H \to M^H \to M$$

Hence it is H-invariant. Picking right actions, we get:

$$\overline{\delta(gh)} = \overline{\delta}(g) \qquad \forall h \in H$$

Therefore, $gm - m = ghm - m$ for all $g \in G$, $h \in H$. In particular, applying this to $g = 1$, then the left hand side is zero, therefore $hm = m$, and so $m \in M^H$. But then δ is a coboundary, and hence the inflation map is injective.

Now let's show that the kernel of the restriction is the image of the inflation. Let $\delta : G \to M$ be a 1-cocycle whose restriction to H is zero (i.e. A 1-coboundary). Then there exists an $m \in M$ such that $\delta(h) = hm - m$ for all $h \in H$. Using this m, define the map $\Delta : G \to M$ where $\Delta(g) = gm - m$. Consider $\varphi = \delta - \Delta$. By construction $\varphi|_H = 0$. Furthermore, φ is a 1-cocycle since δ, Δ are both 1-cocycles (Δ is a 1-coboundary, making it a 1-cocycle). Therefore, $\varphi(g_1 g_2) = \varphi(g_1) + g_1\varphi(g_2)$. Now taking $g_2 \in H$ and use the fact that $\varphi|_H = 0$ so that $\varphi(g_2) = 0$, then

$$\varphi(gh) = \varphi(g)$$

Thus, φ is well-defined on G/H. Let's now show it takes values in M^H, and indeed:

$$\varphi(g) = \varphi(hg) = \varphi(h) + h\varphi(g) = 0 + h\varphi(g) = h\varphi(g)$$

showing it is H-invariant, hence the image is in M^H. Thus:

$$\ker \mathrm{Res} \subseteq \mathrm{imInf}$$

Conversely, if δ is a 1-cocycle in $H^1(G, M)$ that comes from inflation, its restriction to H is constant (by definition of inflation with the $\bmod H$) and it is easy to see using the derivation rule that:

$$\delta(k) = 0 \qquad k \text{ a constant}$$

giving exactness, as we sought to show.

The next two parts of the exact sequence can be shown to be:

$$1 \to H^1(G/H, M^H) \to H^1(G, M) \to H^1(H, M) \to H^2(G/H, M^H) \to H^2(G, M)$$

But to continue one requires the Lyndon–Hochschild–Serre spectral sequence.

Exercise 0.1.1

1. Let A, B be any G-modules. Show that $H^n(G, A \oplus B) \cong H^n(G, A) \oplus H^n(G, B)$
2. Show that $H^0(G, \mathrm{CoInd}^G(A)) \cong A$ and $H^n(G, \mathrm{CoInd}^G(A)) \cong 0$ for all $n \geq 1$ (hint: consider $\zeta : \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}[G^n], \mathrm{CoInd}^G(S)) \to \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G^n], A)$ given by $\varphi \mapsto (z \mapsto \varphi(z)(1))$. What is the inverse of this map?)
3. Show that $H_0(G, \mathrm{Ind}^G(A)) \cong A$ and $H_n(G, \mathrm{Ind}^G(A)) \cong 0$ for all $n \geq 1$.

1

Galois Cohomology

let L/K be a finite extension of local number fields (note that our number fields have finite residue). Let $\mathcal{O}_L^* = U_L$ and let π_L be a uniformizer for L. Define:

$$U_L^i = \left\{ x \in \mathcal{O}_L^* \;\middle|\; x \equiv 1 \bmod \pi_L^i \right\} = 1 + \pi^i \mathcal{O}_L$$

Recall that $\mathcal{O}_L^* = \varprojlim \frac{\mathcal{O}_L^*}{U_L^i}$. Then there is a natural filtration:

$$\mathcal{O}_L^* = U_L^0 \supseteq U_L^1 \supseteq U_L^2 \supseteq \cdots$$

Similarly, by letting $G = \mathrm{Gal}(L/K)$ define the higher ramification groups:

$$G_i = \left\{ g \in G \;\middle|\; gx \equiv x \bmod \mathfrak{m}_L^{i+1}, \text{ for all } x \in \mathcal{O}_L \right\}$$

Then there is a filtration:

$$G = G_{-1} \supseteq G_0 \supseteq G_1 \supseteq \cdots$$

Note that this filtration is eventually 0 since $\mathcal{O}_L/\mathfrak{m}$ is finite. Note that each $G_i \trianglelefteq G$ ("obviously" because the galois action preserves $\mathfrak{m}_L^{i+1}$) and:

$$\frac{G_{-1}}{G_0} \cong \mathrm{Gal}(\mathbb{F}_L/\mathbb{F}_K)$$

where $\mathbb{F}_L, \mathbb{F}_K$ are the residue fields. A very important use of these groups is in another definition of the *different* $\mathcal{D}_{L/K}$ where:

$$\nu_L(\mathcal{D}_{L/K}) = \sum_{i=0}^{\infty} (|G_i| - 1)$$

which is a very interesting connection between the ramification and some G_i-invariance. Note too that G_i/G_{i+1} is the product of cyclic groups of over $p = \nu_L(\mathfrak{m}_p)$, but G_1 is not necessarily abelian (it is p-group).

(I think I should in my CFT notes show *why* abeleaness naturally is to be studied in the context of number theory through this info!)

Let us now link these two filtrations. Let $g \in G_i$. Then:

$$\frac{g(\pi_L)}{g} \in U_L^i$$

and the converse certainly holds. Using these, we can define the map

$$G \to U_L (= \mathcal{O}_L^*) \qquad g \mapsto \frac{g(\pi_L)}{g}$$

which is well-defined as shown by the equation above and by the fact that G, U_L have the aforementioned filtrations. This map depends on the uniformizer π_L,

Proposition 1.0.1: Cocycle of $H^1(G, \mathcal{O}_L^*)$

let f_{π_L} be as above. Then:

$$[f_{\pi_L}] \in H^1(\mathrm{Gal}(L/K), \mathcal{O}_L^*)$$

***Proof*:**

$$\begin{aligned} f_{\pi_L}(gh) = \frac{gh\pi_L}{\pi_L} &= \frac{gh \cdot \pi_L}{g \cdot \pi_L} \frac{g \cdot \pi_L}{\pi_L} \\ &= (g \cdot f_{\pi_L}(h))\, f_{\pi_L}(g) \end{aligned}$$

Now if we change π_L by unit u, then:

$$\frac{g \cdot (\pi_L u)}{\pi_L u} = \frac{(g \cdot \pi_L)(g \cdot u}{\pi_L u}$$

But then we $u \mapsto \frac{g \cdot u}{u}$ is a coboundary, hence it is trivial in cohomology.

Now, consider the following extra quotient mp to f_{π_L} (where the domain is restricted):

$$G_i \xrightarrow{f_{\pi_L}} U_L^i \twoheadrightarrow \frac{U_L^i}{U_L^{i+1}}$$

Call the composition θ_i. Now by definition of f_{π_L}, the image of a $g_{i+1} \in G_{i+1}$ would be in the kernel of θ_i as $f_{\pi_L}(g_{i+1}) \in U_L^{i+1}$. Thus, using the same notation, we have the map:

$$\theta_i : \frac{G_i}{G_{i+1}} \to \frac{U_L^i}{U_L^{i+1}}$$

Proposition 1.0.2: Properties of Theta Map

1. θ_i is injective
2. θ_i does not depend on the choice of uniformizer

***Proof*:**

1. Consider $\theta_i : G_i \to \frac{U_L^i}{U_L^{i+1}}$, it is enough to show the kernel is G_{i+1}. Certainly $G_{i+1} \in \ker\theta_i$. If $g \in \ker\theta_i$ ($g \in G_i$ so $g \cdot \pi_L \equiv \pi_L \mod (\pi_L^{i+1})$) so that $\theta_i(g) = 1$, then:
$$\frac{g \cdot \pi_L}{\pi_L} \equiv 1 \mod \pi_L^{i+1} \qquad \text{i.e.} \qquad g \cdot \pi_L \equiv \pi_L \mod \pi_L^{i+1}$$
Now . In particular, $\frac{g\cdot\pi_L}{\pi_L} \in 1 + \alpha\pi_L^{i+1}$, so it goes to $1 \in U_L^{i+1}$ so so π_L must divide α. Thus we see that g satisfies:
$$g \cdot \pi_L \equiv \pi_l \mod (\pi_L^{i+2})$$
and hence $g \in G_{i+1}$.
2. follows from G_i/G_{i+1}being abelian (as U_L is abelian) (exercise).

Now, what is U_L^i/U_L^{i+1}. First off, the map
$$\mathcal{O}_L^* \to \mathbb{F}_L^* \qquad u \mapsto u \mod \mathfrak{m}_L$$
shows that:
$$\frac{U_L}{U_L^1} = \frac{\mathcal{O}_L^*}{1 + \pi_L\mathcal{O}_L} = \mathbb{F}_L^*$$
where the later is cyclic. For the rest, I'm not sure why I'll get back to this
$$\frac{U_L^i}{U_L^{i+1}} \cong \mathbb{F}_L$$
(which also shows it's an elementary p-group because we know the structure of the finite field $\mathbb{F}_L$)

Proposition 1.0.3: Structure of $H^1(\mathrm{Gal}(L/K), \mathcal{O}_L^*)$

The group $H^1(\mathrm{Gal}(L/K), \mathcal{O}_L^*)$ is cyclic and generated by f_{π_L}.

***Proof*:**
https://youtu.be/UhdlZAJYOw4?t=5073

Index

Bibliography

[1] Nathanael Chwojko-Srawley. *Everything You Need To Know About Elliptic Curves*. 2025. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_elliptic_curves.pdf.

[2] Nathanael Chwojko-Srawley. *Everything You Need To Know About Algebraic Topology*. 1st ed. 2022. URL: https://nathanaelsrawley.com/assets/pdfs/notes/EYNTKA_algebraic_topology.pdf.